OS-Specific VPN Integration APIs & Requirements — Deep Research

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> **Editorial note (added during the 2026-07-04 MVP2 gap-analysis/hardening pass):** raw research brief preserved as historical input, not a living spec. Note in particular that OpenVPN/IKEv2 are covered here as candidate protocols, but the final architecture (`../MVP2_SHARED_CORE.md` §1.5/§2.3) ships WireGuard/Shadowsocks/MASQUE/Multi-Hop only — OpenVPN is a reserved, unimplemented `ProtocolType` enum placeholder and IKEv2 was not carried forward at all. Where this brief's recommendations differ from `../MVP2_ARCHITECTURE.md` / `../MVP2_SHARED_CORE.md`, the specification documents are authoritative.

Research Date: July 2025
Scope: Native VPN APIs, network extension frameworks, and system-level integration requirements for designing Rust shared core platform adapters
Protocols Covered: WireGuard, OpenVPN, IKEv2/IPsec
Platforms: macOS/iOS, Android, Windows, Linux, HarmonyOS, Aurora OS (Sailfish OS)
Searches Performed: 12 independent web search queries across 60+ sources

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1. Executive Summary

This research provides a comprehensive analysis of the native VPN integration APIs across six operating system platforms, with focus on designing platform adapters for a Rust-based VPN shared core. Each OS presents a fundamentally different architecture for VPN integration, with varying capabilities, limitations, and certification requirements.

Key Findings at a Glance

Platform	Primary API	Custom Protocol Support	Kill Switch Support	Split Tunneling
macOS/iOS	NetworkExtension (NEPacketTunnelProvider)	Yes (packet-level)	PF firewall (macOS), Limited (iOS)	Per-app via includedRout excludedRou
Android	VpnService + Builder	Yes (IP packet tunnel)	Always-on + lockdown mode (API 24+)	Per-app (allowed/ disallowed apps, routes)
Windows	WFP callouts + WinTUN/ NDIS	Yes (via WinTUN driver)	WFP firewall rules	IP-based routing
Linux	TUN/TAP + Netlink/ NetworkManager	Full (raw TUN device)	nftables/ iptables	Route-based cgroup-bpf
HarmonyOS				

Platform	Primary API	Custom Protocol Support	Kill Switch Support	Split Tunneling
	VpnExtensionAbility + @ohos.net.vpn	Limited (via TUN fd)	Built-in isBlocking flag	Per-app (trusted/ blocked apps
Aurora/ Sailfish	ConnMan VPN plugin + D-Bus	Via wg-quick/ userspace	iptables/ nftables	Route-based

2. macOS / iOS — NetworkExtension Framework

2.1 Architecture Overview

Apple's NetworkExtension framework is the sole sanctioned mechanism for building VPN applications on macOS and iOS. It provides deep system integration through a privileged app extension model.

The framework distinguishes between two main VPN approaches [^64^]:

> **Personal VPN** (`NEVPNManager`): For standard VPN connections using IPsec or IKEv2 protocols. Leverages Apple's built-in VPN client infrastructure. Suitable for consumer-facing VPN services.
>
> **Network Extensions** (`NETunnelProviderManager`): For implementing custom VPN protocols (WireGuard, OpenVPN, etc.) that are not IPsec/IKEv2. Uses `NEPacketTunnelProvider` for packet-level tunneling or `NEAppProxyProvider` for flow-based proxying.

2.2 Core Classes

Class	Purpose	Platform
`NEPacketTunnelProvider`	Main tunnel logic — receives raw IP packets, manages virtual interface	macOS/ iOS
`NETunnelProviderManager`	Manages VPN configuration in system preferences	macOS/ iOS
`NETunnelProviderProtocol`		macOS/ iOS

Class	Purpose	Platform
	Defines protocol-specific settings (server address, provider bundle ID)	
`NEVPNManager`	Personal VPN management for IPsec/IKEv2 only	macOS/iOS
`NEOnDemandRule`	Rules for automatic VPN activation	macOS/iOS

2.3 NEPacketTunnelProvider Lifecycle

The `NEPacketTunnelProvider` lifecycle follows a well-defined pattern [^142^]:

```
class PacketTunnelProvider: NEPacketTunnelProvider {
    override func startTunnel(options: [String : NSObject]?,
                              completionHandler: @escaping
        (Error?) -> Void)
    override func stopTunnel(with reason: NEProviderStopReason,
                             completionHandler: @escaping () ->
        Void)
    override func handleAppMessage(_ messageData: Data,
                                   completionHandler: ((Data?) ->
        Void)?)
    override func sleep(completionHandler: @escaping () -> Void)
    override func wake()
}
```

Key lifecycle characteristics [^142^]:

- `startTunnel()`: Called when the main app invokes `startVPNTunnel()`. Must call `completionHandler(nil)` on success or `completionHandler(error)` on failure.
- `stopTunnel()`: Called on disconnect. Must call `completionHandler()` when cleanup is complete.
- `cancelTunnelWithError()`: Used internally by the provider to signal an error condition that should trigger reconnection.
- `sleep()`/`wake()`: Critical for handling device sleep/wake transitions. Must restore state properly.
- `handleAppMessage()`: IPC channel between the containing app and the extension via `NETunnelProviderSession.sendProviderMessage()`.

2.4 Entitlements & Provisioning

The Network Extension entitlement (`com.apple.developer.networking.networkextension`) is required [^70^]:

> "Since November 2016, Packet Tunnel, App Proxy, Content Filter and DNS Proxy providers are self-serve — you enable the capability in Xcode or on the developer website with no request. Only Hotspot Helper and the NE app push provider remain managed."

```
<key>com.apple.developer.networking.networkextension</key>
<array>
    <string>packet-tunnel-provider</string>
</array>
```

Critical deployment restriction [^64^]: Provider types have different deployment restrictions. Some work only on **managed/supervised devices**. See Apple's TN3134 for details.

2.5 Tunnel Configuration

Configuration is established through `NETunnelProviderManager` [^69^]:

```
let managers = try await
        NETunnelProviderManager.loadAllFromPreferences()
let manager = managers.first ?? NETunnelProviderManager()

let tunnelProtocol = NETunnelProviderProtocol()
tunnelProtocol.providerBundleIdentifier =
        config.providerBundleIdentifier
tunnelProtocol.serverAddress = config.serverAddress
tunnelProtocol.providerConfiguration = ["key": "value"]

manager.localizedDescription = config.displayName
manager.protocolConfiguration = tunnelProtocol
manager.isEnabled = true

try await manager.saveToPreferences()
```

Important: VPN configurations created using `NETunnelProviderManager` are classified as **enterprise VPN configurations**. Only one enterprise VPN can be enabled at a time. Enterprise VPN takes precedence over Personal VPN when routes conflict [^65^].

2.6 Packet Flow & Transport

`NEPacketTunnelProvider` provides two transport mechanisms for the tunnel [^129^]:

- **UDP**: `createUDPSession(to:)` — preferred for most VPN protocols (faster, lower overhead)
- **TCP**: `createTCPConnection(to:)` — useful for protocols requiring reliability or for traversing restrictive networks

> "The primary API for implementing your custom tunneling protocols in `NEPacketTunnelProvider` allows you to tunnel traffic on an IP layer. They run as app extensions in the background handling network traffic." [^129^]

Packets flow through a virtual `utunX` interface. The system diverts IP packets to the `NEPacketTunnelProvider`, which encapsulates them and sends via UDP/TCP to the VPN server.

2.7 On-Demand Rules

Automatic VPN activation is supported via `NEOnDemandRule` subclasses [^64^]:

```
let onDemandRule = NEOnDemandRuleConnect()
onDemandRule.interfaceTypeMatch = .wiFi
onDemandRule.ssidMatch = ["OfficeNetwork"]

manager.onDemandRules = [onDemandRule]
manager.isOnDemandEnabled = true
```

Also supports `NEOnDemandRuleEvaluateConnection` with `NEEvaluateConnectionRule` for domain-based triggering.

2.8 macOS-Specific: Kill Switch via PF

On macOS, kill switch is implemented using the **Packet Filter (PF)** firewall [^114^] [^219^]:

> "IVPN has implemented a secure and robust mechanism called the IVPN firewall. Once enabled the IVPN Firewall integrates deep into the operating system (using Microsoft's own WFP API on Windows, `pf` on macOS, and `iptables` on Linux) and filters all network packets."

macOS 14+ PF anchors with block/allow policies:

```
block drop all
pass on utunX all
```

WWDC 2025 guidance [^136^]: Apple explicitly discourages using Packet Filter or directly modifying routing tables on Mac. Network Extension is the supported API:

> "Avoid using Packet Filter or directly modifying the routing table on the Mac. This is not supported and risk clashing with traffic filtering and routing rules installed by the system or other apps. If your VPN app doesn't use Network Extension today, you should migrate as soon as possible."

2.9 Key Limitations

Limitation	Details
Only one enterprise VPN active	System-wide limitation; conflicts with other VPN apps
iOS Simulator unsupported	Must test on physical devices
Extension memory limits	System may terminate under extreme memory pressure
No direct routing table modification (macOS)	Must use Network Extension APIs
`NEAppProxyProvider` requires supervised device	Per-app proxy only for managed devices
utun interface number changes	Must handle dynamic interface numbering after sleep/wake
iOS kill switch limited	No true kernel-level kill switch; relies on `disconnectOnSleep` and profile behavior

3. Android — VpnService API

3.1 Architecture Overview

Android's `VpnService` is the core API for building custom VPN clients. It operates by creating a virtual network interface (TUN) and intercepting all IP traffic from the device.

3.2 Core API: VpnService.Builder

The `VpnService.Builder` class configures the VPN tunnel [^109^]:

```
public class VpnService.Builder {
    public Builder addAddress(String address, int prefixLength)
    public Builder addRoute(String address, int
        prefixLength)      // Route through VPN
    public Builder excludeRoute(IpPrefix
        prefix)                    // Exclude from VPN (API 33+)
    public Builder addDnsServer(String address)
    public Builder addSearchDomain(String domain)
    public Builder addAllowedApplication(String
        packageName)        // Per-app VPN (include)
    public Builder addDisallowedApplication(String
        packageName)     // Per-app VPN (exclude)
```

```
    public Builder
        allowBypass()                                    // Allow
        apps to bypass VPN
    public Builder setBlocking(boolean blocking)
    public Builder setMtu(int mtu)
    public Builder setSession(String session)
    public ParcelFileDescriptor
        establish()                          // Create TUN
        interface
}
```

3.3 Always-On VPN & Lockdown Mode

Android 7.0+ (API 24) introduced **Always-on VPN** [^29^]:

> "Android can start a VPN service when the device boots and keep it running. This feature is called *always-on VPN*."

Key characteristics:

- System starts/stops the VPN service automatically
- Persists across reboots and app updates
- VPN app can opt-out via `SERVICE_META_DATA_SUPPORTS_ALWAYS_ON` metadata set to `false`

Lockdown mode (`BLOCK_CONNECTIONS_WITHOUT_VPN`) [^26^]:

> "When `lockdownEnabled` is `true`, all network traffic is blocked if the VPN is not connected. No traffic can leak to the open internet."

Feature	API Level	Behavior
Always-on VPN	24+	Auto-starts VPN on boot, keeps it running
Block connections without VPN	24+	System-level kill switch — blocks all non-VPN traffic
Lockdown exemptions	29+	Specific apps can be exempted from lockdown
User can't disable always-on	30+	Admin-configured always-on cannot be disabled by user
System VPN app exclusion	34+	System apps can be excluded from VPN

3.4 Split Tunneling

Android supports split tunneling through multiple mechanisms [^109^] [^21^]:

Route-based (all Android versions):

```
builder.addRoute("0.0.0.0", 0)  // Route all traffic through VPN
```

Exclude routes (API 33+):

```
builder.excludeRoute(IpPrefix("192.168.1.0", 24))  // Exclude LAN
```

Per-app VPN:

```
builder.addAllowedApplication("com.example.app")      // Only this
        app uses VPN
// OR
builder.addDisallowedApplication("com.example.app")   // This app
        bypasses VPN
```

> "`addAllowedApplication()` and `addDisallowedApplication()` are mutually exclusive — you allowlist specific apps or blocklist them, not both." [^26^]

Pre-API 33 workaround for exclude routes [^21^]:

> "For Android 33+, we used `builder.excludeRoute` to exclude the desired IPs from the VPN. For versions below Android 33, we relied on `builder.addRoute` and included the required IP addresses calculated using the WireGuard AllowedIPs Calculator."

3.5 Background Execution & Foreground Services

VPN services must run as **foreground services** to avoid being killed [^175^]:

> "Due to Android battery optimizations introduced in Android 8.0 (API level 26), background services have now some important limitations. Essentially, they are killed once the app is in background for a while."

Critical Android 15 change [^177^]:

> "Starting in Android 15, all of an app's foreground services share a **6-hour time limit**."

The recommended approach for long-running VPN services:

- Use `dataSync` foreground service type (required for Android 14+)
- Use **User-Initiated Data Transfer (UIDT) Jobs** for Android 14+ to bypass restrictions
- For Android 13 and below: rely on Foreground Services + WorkManager

```
<service android:name=".VpnService"
         android:foregroundServiceType="dataSync"
         android:permission="android.permission.BIND_VPN_SERVICE">
    <intent-filter>
        <action android:name="android.net.VpnService"/>
    </intent-filter>
</service>
```

3.6 Battery Optimization

Enterprise environments can exempt VPN apps from battery optimization via [^63^]:

- Knox Service Plugin `BatteryOptimizationAllowlist` policy
- `addPackageToBatteryOptimizationWhiteList()` API (Knox SDK)
- EMM-managed configuration

3.7 Protecting Sockets

The `VpnService.protect()` method allows specific sockets to bypass the VPN tunnel:

```
val socket = DatagramSocket()
VpnService.protect(socket)  // This socket goes directly to
        physical network
```

This is essential for the VPN's own transport connection to avoid routing loops.

3.8 Key Limitations

Limitation	Details
No raw IP socket access	Must use VpnService TUN interface
Multiple VPNs can't run simultaneously	Only one active VPN at a time
Foreground service notification required	User-visible notification mandatory
6-hour limit on foreground services (Android 15)	Requires UIDT jobs for long-running VPNs
`excludeRoute()` only API 33+	Requires workaround on older versions
Battery optimization may kill service	User or admin must whitelist app
`HttpProxy` over split tunnel doesn't work	Proxy can't reach excluded destinations

4. Windows — Filtering Platform & NDIS

4.1 Architecture Overview

Windows provides multiple APIs for VPN integration, ranging from user-mode UWP plug-ins to kernel-mode WFP callout drivers.

4.2 Primary APIs

API Layer	Technology	Use Case
Windows Filtering Platform (WFP)	Kernel-mode callout drivers	Kill switch, traffic filtering, packet inspection
NDIS Lightweight Filter (LWF)	Kernel-mode driver	Low-level packet interception
UWP VPN Platform	`Windows.Networking.Vpn` namespace	Store-distributed VPN apps
RAS API	User-mode (Win32)	VPN profile management, dial-up
WinTUN / WinTap	Virtual network adapter drivers	TUN/TAP device for packet tunneling

4.3 Windows Filtering Platform (WFP)

WFP is the primary mechanism for implementing kill switches and traffic filtering on Windows [^219^]:

> "On Windows, the core mechanism is the Windows Filtering Platform (WFP). The VPN client installs a callout driver or uses system layers to filter at transport and network stack levels, marking VPN interface traffic and blocking all else."

WFP filtering layers relevant to VPN:

- `FWPM_LAYER_OUTBOUND_IPPACKET_V4` / `V6` — Outbound IP packets
- `FWPM_LAYER_INBOUND_IPPACKET_V4` / `V6` — Inbound IP packets
- `FWPM_LAYER_STREAM_V4` / `V6` — TCP/UDP streams
- `FWPM_LAYER_IPFORWARD_V4` / `V6` — Forwarding decisions

Kill switch implementation via WFP [^221^]:

> "On Windows — Use WFP to implement a block everything rule, then provide a higher priority rule to allow on the tunnel interface."

4.4 WinTUN Driver

The WireGuard project provides **WinTUN**, a virtual TUN adapter for Windows:

> "If an interface has only one peer, and that peer contains an Allowed IP in `/0`, then WireGuard enables a so-called 'kill-switch', which adds firewall rules to do the following: Packets from the tunnel service itself are permitted; Loopback packets are permitted; DHCP for IPv4 and IPv6 and NDP for IPv6 are permitted; All other packets are blocked." [^156^]

Key behavior [^156^]:

- WinTUN automatically creates kill-switch firewall rules when `AllowedIPs = 0.0.0.0/0, ::/0`
- Uses `0.0.0.0/1` + `128.0.0.0/1` split to avoid triggering kill-switch semantics when not desired
- DNS on port 53 is restricted to configured DNS servers only

4.5 UWP VPN Plug-in Framework

Microsoft provides a UWP VPN platform for Store apps [^199^]:

> "The Windows UWP VPN platform handles all aspects of creating a virtual network adapter and adding it to the system along with appropriate metrics and Name Resolution Policy Table (NRPT) entries, DNS servers, and more."

Key features:

- System tray integration
- Custom XML configuration (MDM/Intune push)
- Web authentication support (`VpnForegroundActivatedEventArgs`)
- Route and traffic filter configuration via `StartWithTrafficFilter`
- Packet encapsulation/decapsulation in background task

4.6 NDIS vs WFP

Community guidance from OSR Developer Community [^73^]:

> "On NT6 use WFP unless your packet processing requires you to deal with MAC layer frames. Since you mention VPN and the only transport left is IP, I presume you mean an IPv4/6 VPN. So WFP will be just what you want."

Modern recommendation: For Windows 10/11, use WFP exclusively to maintain compatibility with HVCI (Hypervisor-Protected Code Integrity).

4.7 Kill Switch Implementation

Professional VPN clients implement kill switches on Windows using [^219^]:

```
# Block all outbound
New-NetFirewallRule -Direction Outbound -Action Block -Profile Any

# Allow VPN interface
New-NetFirewallRule -Direction Outbound -Action Allow
        -InterfaceAlias "WireGuard Tunnel"

# Set interface metrics
Set-NetIPInterface -InterfaceAlias "VPN" -InterfaceMetric 1
```

Advanced clients use WFP callouts for faster, more reliable filtering.

4.8 Key Limitations

Limitation	Details
WFP callout drivers require kernel-mode code	Must be signed, complex development
BFE (Base Filtering Engine) service restart	Firewall rules may be lost; need restoration logic
Multiple filter drivers can conflict	Need careful weight/layer positioning
NDIS LWF drivers also require signing	WHQL certification recommended
Interface metric management	Windows auto-metric can override VPN priority

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5. Linux — TUN/TAP & NetworkManager

5.1 Architecture Overview

Linux offers the most flexible and open VPN integration environment. The standard approach uses the kernel's TUN/TAP virtual network interface driver combined with userspace management tools.

5.2 TUN/TAP Device Creation

TUN devices are created via the `/dev/net/tun` character device:

```
int fd = open("/dev/net/tun", O_RDWR);
struct ifreq ifr;
memset(&ifr, 0, sizeof(ifr));
ifr.ifr_flags = IFF_TUN | IFF_NO_PI;
strcpy(ifr.ifr_name, "tun0");
ioctl(fd, TUNSETIFF, (void *)&ifr);
```

Required capabilities: `CAP_NET_ADMIN` for creating TUN devices. Can be granted via file capabilities:

```
sudo setcap cap_net_admin,cap_net_raw=eip ./vpn-client
```

5.3 NetworkManager Integration

NetworkManager 1.16+ provides native WireGuard support [^67^]:

> "NetworkManager provides a de facto standard API for configuring networking on the host. This allows different tools to integrate and interoperate — from cli, tui, GUI, to cockpit."

D-Bus API for VPN management:

```
# Create WireGuard connection
nmcli connection add type wireguard ifname wg0 con-name my-wg0

# Configure and activate
nmcli connection modify my-wg0 ipv4.method manual ipv4.addresses
        192.168.7.5/24
nmcli --show-secrets --ask connection up my-wg0
```

NetworkManager D-Bus interfaces for VPN:

- `org.freedesktop.NetworkManager.Device.WireGuard` — WireGuard device operations
- VPN plugin service files in `/usr/lib/NetworkManager/VPN/`

5.4 WireGuard Kernel Module

WireGuard has been in the Linux kernel since 5.6 [^71^]:

```
/lib/modules/5.10.6-1/kernel/drivers/net/wireguard/wireguard.ko.gz
```

This provides native, high-performance WireGuard tunneling without userspace implementations.

5.5 systemd-networkd Integration

Alternative to NetworkManager, `systemd-networkd` can manage VPN interfaces via `.netdev` and `.network` files:

```
# /etc/systemd/network/wg0.netdev
[NetDev]
Name=wg0
Kind=wireguard

[WireGuard]
PrivateKey=<base64-private-key>
ListenPort=51820

[WireGuardPeer]
PublicKey=<peer-public-key>
AllowedIPs=0.0.0.0/0,::/0
Endpoint=<server>:51820
```

5.6 Kill Switch Implementation (nftables)

Modern Linux kill switches use **nftables** (kernel 5.10+ recommended) [^219^]:

```
table inet vpn {
    chain output {
        type filter hook output priority 0; policy drop;
        oifname "wg0" accept
        oifname "lo" accept
        ct state established,related accept
        ip daddr <vpn-server-ip> accept  # Allow VPN endpoint
        drop
    }
}
```

Policy routing with fwmark:

```
# Mark VPN traffic
ip rule add fwmark 0x1 table 100
ip route add default dev wg0 table 100
```

5.7 Split Tunneling (cgroup-bpf)

Advanced split tunneling uses **cgroup-bpf** for process-level filtering [^219^]:

> "For advanced setups, cgroup-bpf (BPF_CGROUP_INET_EGRESS) filters at the process level — letting, for example, your admin's ssh bypass the kill switch for debugging while everything else remains blocked."

5.8 Flatpak/Snap Sandboxing Implications

Sandboxed Linux app distributions present challenges for VPN clients:

Sandboxing System	Network Access	TUN Device	Workaround
Flatpak	Portal-based (`NetworkMonitor`)	No direct access	Use `--device=all` or run outside sandbox
Snap	`network`, `network-bind` plugs only	No TUN access	Classic confinement or snapd plugin

VPN clients distributed via Flatpak/Snap typically cannot create TUN devices due to sandbox restrictions. Common approaches:

- Distribute the UI via Flatpak, run the VPN daemon as a systemd service outside the sandbox
- Use `pkexec` for privilege escalation
- Request classic confinement (Snap) or `--device=all` (Flatpak)

5.9 Key Limitations

Limitation	Details
Requires root or CAP_NET_ADMIN	Must grant capabilities
Multiple VPN management systems	NetworkManager vs systemd-networkd vs manual scripts
iptables vs nftables fragmentation	iptables is deprecated; nftables is modern but not universal
No unified sandbox support	Flatpak/Snap break TUN access
DNS leak prevention is manual	Must configure systemd-resolved or resolv.conf management

6. HarmonyOS — VpnExtensionAbility

6.1 Architecture Overview

HarmonyOS provides a VPN framework through `VpnExtensionAbility` and the `@ohos.net.vpn` module. The architecture follows a model similar to Android's VpnService but with HarmonyOS-specific APIs.

6.2 VpnExtensionAbility

The `VpnExtensionAbility` class provides lifecycle callbacks for third-party VPNs [^77^]:

```
import { VpnExtensionAbility } from '@kit.NetworkKit';
import { Want } from '@kit.AbilityKit';

class MyVpnExtAbility extends VpnExtensionAbility {
    onCreate(want: Want) {
        console.info('MyVpnExtAbility onCreate');
    }
    onDestroy() {
        console.info('MyVpnExtAbility onDestroy');
    }
}
```

Context: `VpnExtensionContext` provides the VPN-specific runtime context.

6.3 VpnConnection API

The `@ohos.net.vpn` module provides `VpnConnection` for VPN management [^195^]:

```
import vpn from '@ohos.net.vpn';

// Create VPN connection
const VpnConnection = vpn.createVpnConnection(context);

// Set up VPN (returns TUN file descriptor)
let config: vpn.VpnConfig = {
    addresses: [{
        address: { address: "10.0.0.5", family: 1 },
        prefixLength: 24
    }],
    mtu: 1400,
    dnsAddresses: ["114.114.114.114"],
    isBlocking: true,        // Kill switch
    trustedApplications: [],  // Per-app split tunneling (include)
    blockedApplications: []   // Per-app split tunneling (exclude)
};

VpnConnection.setUp(config).then((tunfd: number) => {
    console.info("setUp success, tunfd: " + tunfd);
});

// Protect socket (bypass VPN)
VpnConnection.protect(socketFd);
```

```
// Destroy VPN
VpnConnection.destroy();
```

6.4 Permission Model

VPN functionality requires the `ohos.permission.MANAGE_VPN` permission [^197^]:

Attribute	Value
Permission	`ohos.permission.MANAGE_VPN`
Permission level	`system_basic`
Authorization mode	`system_grant`
Enable via ACL	true (API 12+)
Valid since	API 10

> "Allows a system application to enable or disable the VPN function." [^197^]

API 10-11: Only system apps could use VPN APIs.
API 12+: ACL (Access Control List) enables third-party apps to request this permission.

6.5 VpnConfig Options

Field	Type	Required	Description
`addresses`	`Array<LinkAddress>`	Yes	IP address of the vNIC
`routes`	`Array<RouteInfo>`	No	Route information
`dnsAddresses`	`Array<string>`	No	DNS server IPs
`mtu`	`number`	No	Maximum transmission unit
`isIPv4Accepted`	`boolean`	No	IPv4 support (default: true)
`isIPv6Accepted`	`boolean`	No	IPv6 support (default: false)
`isBlocking`	`boolean`	No	Blocking mode / kill switch (default: false)
`trustedApplications`	`Array<string>`	No	

Field	Type	Required	Description
			Bundle names of trusted apps
`blockedApplications`	`Array<string>`	No	Bundle names of blocked apps
`isLegacy`	`boolean`	No	Built-in VPN support (default: false)

6.6 Key Limitations

Limitation	Details
`MANAGE_VPN` is system-level permission	Requires system app or ACL (API 12+)
No direct raw socket access	Must use provided socket APIs
Third-party VPN is relatively new	Ecosystem still maturing
ArkTS/ArkUI native C interop	May need NAPI for Rust integration

7. Aurora OS / Sailfish OS — ConnMan Integration

7.1 Architecture Overview

Aurora OS (Russian Sailfish OS fork) uses the same networking stack as Sailfish OS, centered on **ConnMan** (Connection Manager) as the system network manager. VPN integration is achieved through ConnMan VPN plugins.

7.2 ConnMan VPN Plugin Architecture

Sailfish OS uses ConnMan's VPN plugin system for VPN integration [^152^]:

> "SailfishOS doesn't provide WireGuard functionality out-of-the-box, so we first need to install a few third-party programs from OpenRepos."

Required packages for WireGuard on Sailfish OS:

- `wireguard-go` — Userspace WireGuard implementation
- `wireguard-tools` — Configuration utilities
- `connman-plugin-vpn-wireguard` — ConnMan VPN plugin
- `jolla-settings-networking-plugin-vpn-wireguard` — Settings UI integration

7.3 VPN Plugin Structure

ConnMan VPN plugins communicate via **D-Bus** [^153^]:

```
net.connman.vpn  — D-Bus service for VPN management
net.connman.vpn.Connection  — Individual VPN connection interface
net.connman.vpn.Manager  — VPN manager interface
```

Key D-Bus properties:

- `State` — Connection state (`association`, `configuration`, `ready`, `failure`)
- `SplitRouting` — Split tunneling enabled/disabled

7.4 Configuration

Sailfish OS 5.0+ includes built-in WireGuard support [^154^]:

```
devel-su pkcon install jolla-settings-networking-plugin-vpn-
        wireguard
systemctl restart connman
```

Configuration is done through the Settings > System > VPN UI, or by importing WireGuard configuration files.

7.5 Key Characteristics

Aspect	Details
Network manager	ConnMan
VPN plugin API	ConnMan VPN D-Bus API
Settings integration	Jolla Settings VPN plugins
DNS handling	`connmand` handles all DNS (127.0.0.1) [^154^]
Userspace fallback	`wireguard-go` when kernel module unavailable [^145^]
Platform SDK	Qt/QML based

7.6 Key Limitations

Limitation	Details
No kernel WireGuard module	Uses userspace `wireguard-go`
Third-party package ecosystem	Relies on OpenRepos community packages
ConnMan crashes reported	Stability issues with VPN plugins [^153^]
Limited native development docs	Community documentation only

8. Permission Models & MDM Integration

8.1 iOS/macOS Permissions & MDM

Aspect	Details
App Store Review	VPN apps face strict review; must justify Network Extension usage
Entitlement	`com.apple.developer.networking.networkextension` (self-serve since 2016)
Supervised devices	Some features (`NEAppProxyProvider`, content filtering) require supervised mode
MDM configuration	`com.apple.vpn.managed` payload for profile-based VPN
Per-app VPN	Supported via MDM for managed apps

Enterprise VPN deployment [^70^]:

- VPN configurations pushed via Apple Configurator, Profile Manager, or third-party MDM
- On-demand rules can be pre-configured
- Personal VPN vs Enterprise VPN precedence rules apply

8.2 Android Permissions & MDM

Aspect	Details
Permission	`android.permission.BIND_VPN_SERVICE`
System dialog	User must grant VPN permission via system dialog (one-time per app)
	Fully managed, work profile, COPE modes

Aspect	Details
MDM (Android Enterprise)	
Always-on VPN	Configurable via MDM; cannot be disabled by user (API 30+)
Lockdown mode	Admin-configurable; blocks all non-VPN traffic
Per-app VPN	Via `addAllowedApplication()` / `addDisallowedApplication()`
Battery optimization	Admin can whitelist via EMM policies

Key MDM policies [^26^]:

- `DISALLOW_CONFIG_VPN` — Prevent user from changing VPN
- `vpnConfigDisabled` — Disable VPN configuration UI
- `lockdownEnabled` — Enable kill switch
- Battery optimization allowlist — Exempt VPN from Doze

8.3 Windows MDM Integration

Aspect	Details
MDM Support	Microsoft Intune, VMware Workspace ONE, etc.
VPN Profile	`VPN` configuration service provider (CSP)
Native VPN	Built-in IKEv2 client configurable via MDM
Third-party VPN	UWP VPN plugins can receive MDM config via custom XML [^199^]
WEF (Windows Event Forwarding)	Can monitor VPN events

8.4 Linux MDM/Enterprise

Aspect	Details
NetworkManager	D-Bus API for configuration management
`nmcli`/`nmtui`	CLI/TUI tools for VPN configuration
systemd-networkd	`.netdev`/`.network` file-based configuration
Puppet/Ansible/ Chef	Common for enterprise Linux VPN deployment
No unified MDM	Unlike mobile platforms, no single MDM standard

8.5 HarmonyOS MDM

HarmonyOS supports MDM through the **MDM Kit** [^174^]:

> "Adds support for enabling or disabling the SMS, mobile data, airplane mode, notification, and NFC features. Adds multiple user behavior restriction policies for PCs and 2-in-1 devices."

VPN configuration is part of enterprise device management via Endpoint Central and similar platforms [^176^]:

> "Centrally manage device configurations like Wi-Fi, VPN, email, certificates, and more."

---

9. Kill Switch Implementation

9.1 Cross-Platform Kill Switch Approaches

Platform	Mechanism	Implementation	Reliability
macOS	PF (Packet Filter) firewall	Kernel-level anchor rules; block all except utun	High
iOS	Network Extension profile	Limited; `disconnectOnSleep`, profile-based block	Medium
Android	System lockdown mode	`BLOCK_CONNECTIONS_WITHOUT_VPN` + Always-on VPN	High (system-level)
Windows	WFP firewall rules	Block outbound except VPN interface; WinTUN auto-rules	High
Linux	nftables/ iptables + policy routing	Drop default; allow only tunnel interface	High
HarmonyOS	`isBlocking` in VpnConfig	Built-in blocking mode	Medium

9.2 macOS Kill Switch (PF)

> "On macOS, the kill switch hinges on Network Extension (NE) and the built-in Packet Filter (PF). An NE client controls the utun tunnel interface, while PF anchors enforce strict policies: block all outbound traffic except on utunX and allowed tunnel services." [^219^]

Implementation:

```
block-policy drop
skip on lo0
anchor "vpn-killswitch" {
    block drop all
    pass on utun* all
}
```

9.3 iOS Kill Switch Limitations

iOS does **not** provide a true kernel-level kill switch for third-party VPN apps [^117^]:

> "On iOS, it seems like a kill switch isn't even possible on any version."

Workarounds:

- Use Apple's Always-on VPN (requires MDM/supervised device)
- `NEVPNProtocol.disconnectOnSleep` can disconnect on sleep
- Some apps implement "Network Protection" features that are limited

9.4 Android Kill Switch (Lockdown Mode)

Android's system-level kill switch is the most reliable on mobile [^107^]:

> "Android 8.0 and later includes a native kill switch feature called 'Always-on VPN' combined with 'Block connections without VPN'. This system-level option works with any VPN app and provides reliable protection without depending on the VPN developer's implementation."

```
// Programmatic detection (not control — user/system must enable)
val vpnManager = context.getSystemService(Context.VPN_SERVICE) as
        VpnManager
// Check if always-on is configured for this app
```

9.5 Windows Kill Switch (WFP)

> "The approach: set a Block All outbound rule, then carve out exceptions allowing traffic via the VPN interface. Advanced clients go further — installing WFP callouts that drop packets before TCP stack processing and mark VPN packets to bypass accidental allow rules." [^219^]

WireGuard's built-in kill switch [^156^]:

> "If an interface has only one peer, and that peer contains an Allowed IP in `/0`, then WireGuard enables a so-called 'kill-switch', which adds firewall rules to drop all traffic that is not travelling over the VPN."

9.6 Linux Kill Switch (nftables)

> "By 2026, iptables is still around but outdated. We create an inet table called vpn, with input, forward, and output chains. Output policy is drop. We allow: established/related connections; interface wg0 (or tun0); DNS over wg0; optionally localhost." [^219^]

---

10. Split Tunneling

10.1 Platform Support Summary

Platform	Mechanism	Granularity	Per-App Support
macOS/iOS	`includedRoutes`/ `excludedRoutes`	IP/subnet	Limited (iOS not possible per-app)
Android	`addRoute()`/ `excludeRoute()` + app lists	IP/subnet + app	Yes (`addAllowedApplication`/ `addDisallowedApplication`)
Windows	Route table + WFP filters	IP/subnet + process	Yes (WFP process-based)
Linux	Policy routing + cgroup-bpf	IP/subnet + process	Yes (cgroup-bpf)
HarmonyOS	`trustedApplications`/ `blockedApplications`	App	Yes

10.2 iOS Split Tunneling Limitations

Mullvad documents this clearly [^111^]:

> **"Q: Will you add split tunneling to the iOS / iPadOS app?"**
> **"A: It is not possible to do this currently."**

iOS Network Extension provides `NEIPv4Route` and `NEIPv6Route` for route-based splitting only. Per-app split tunneling is not available for third-party VPNs.

10.3 Android Split Tunneling

Android provides the most comprehensive split tunneling on mobile [^109^]:

Route-based (all versions):

```
builder.addRoute("0.0.0.0", 0)          // Default through VPN
builder.addRoute("10.0.0.0", 8)         // Corporate network
        through VPN
```

Exclude routes (API 33+):

```
builder.excludeRoute(IpPrefix("192.168.0.0", 16))  // Exclude LAN
```

Per-app (all versions):

```
builder.addAllowedApplication("com.company.app")       // Only
        these apps use VPN
builder.addDisallowedApplication("com.local.app")      // These
        apps bypass VPN
```

10.4 Windows Split Tunneling

Windows split tunneling is typically implemented via:

1. **Route table management**: Add/remove routes for specific subnets
2. **WFP callout drivers**: Process-based routing decisions [^224^]

> "A callout driver implementing policy-based routing for Windows, based on process name. Redirects TCP connections of a given process into a given network, despite of a default route." [^224^]

10.5 Linux Split Tunneling

Linux offers the most flexible split tunneling:

- **Route-based**: `ip route` commands for subnet-based routing
- **cgroup-bpf**: Process-level filtering via eBPF

- **Network namespaces**: Complete network isolation per process group

---

11. Background Execution Constraints

11.1 iOS/macOS Background Execution

Network Extension providers run as **separate system processes**, independent of the hosting app [^64^]:

> "Apple has engineered this framework to allow VPN connections to operate at a system level, independent of the lifecycle of the application that configured them."

- The extension is managed by the OS, not the app
- Persists even if the containing app is terminated
- May be terminated under extreme memory pressure
- Must handle `sleep()`/`wake()` events properly

11.2 Android Background Execution

Android Version	Constraint	Impact on VPN
8.0 (API 26)	Background service limits	Must use foreground service
9.0 (API 28)	App Standby Buckets	May restrict background VPN service
10 (API 29)	Background location restrictions	Affects location-aware VPN features
12 (API 31)	Foreground service launch restrictions	Must use notification trampoline
14 (API 34)	Foreground service type required	Must declare `dataSync` type
15 (API 35)	6-hour foreground service limit	Requires UIDT jobs for long-running VPNs

Mitigation strategies [^177^]:

- Foreground service with `dataSync` type
- User-Initiated Data Transfer (UIDT) Jobs (API 34+)
- Battery optimization exemption
- MDM-managed always-on VPN

11.3 Windows Background Execution

VPN on Windows typically runs as a **Windows Service**:

- Can start automatically on boot
- Not subject to user session limitations
- WFP callout drivers are kernel-mode (always active)
- UWP VPN plugins run as background tasks

11.4 Linux Background Execution

VPN on Linux runs as a **systemd service**:

- `systemd-networkd` or NetworkManager manages interfaces
- WireGuard interfaces can be configured to auto-start
- No arbitrary execution time limits
- Persists across user sessions

11.5 HarmonyOS Background Execution

`VpnExtensionAbility` runs as a system ExtensionAbility:

- Lifecycle managed by the Ability runtime
- `onCreate()`/`onDestroy()` callbacks for setup/teardown
- `AppServiceExtensionAbility` (API 12+) for extended background service capabilities [^174^]

---

12. Raw Socket Access

12.1 Platform Comparison

Platform	Raw Socket Access	VPN Relevance	Requirements
Linux	Full support via `socket(AF_PACKET, ...)`	Custom protocol framing	`CAP_NET_RAW` capability
macOS	Supported (root or entitlement)	Low-level packet capture	Root or specific entitlement
iOS	**Not available to third-party apps**	Must use NetworkExtension APIs	N/A
Android	**Not available**	Must use VpnService TUN	N/A
Windows	Supported (admin + driver)	NDIS/WFP development	Administrator privileges
HarmonyOS			

Platform	Raw Socket Access	VPN Relevance	Requirements
	Limited (via socket APIs)	Use provided network APIs	`MANAGE_VPN` permission

12.2 Linux Raw Socket Details

Linux provides full raw socket access with `CAP_NET_RAW` [^143^]:

> "The very handy `CAP_NET_RAW` capability can be used to open raw sockets. Capabilities are applied on a per-file basis with the `setcap` command."

```
sudo setcap cap_net_admin,cap_net_raw=eip ./vpn-client
```

Required capabilities for VPN [^147^]:

- `CAP_NET_ADMIN` — Network interface configuration, routing tables
- `CAP_NET_RAW` — Raw and packet sockets

12.3 Mobile Platform Limitations

Both iOS and Android **do not** provide raw socket access to third-party apps:

- iOS: Must use `NEPacketTunnelProvider` with TUN abstraction
- Android: Must use `VpnService.Builder.establish()` to get TUN fd
- Custom protocol implementations must work over the provided TUN interface

12.4 Implications for Rust Shared Core

The Rust shared core should:

- Accept a TUN file descriptor (or platform equivalent) as the packet I/O channel
- Not attempt raw socket operations on mobile platforms
- Use platform-provided transport mechanisms (UDP/TCP sockets) for the VPN tunnel connection

---

13. Network Extension Lifecycle

13.1 iOS/macOS Lifecycle

```
[App Saves Config] -> [User Authorizes] -> [System Manages
Extension]
```

```
                    |
                    v
            [NEPacketTunnelProvider]
                    |
      +-------------+-------------+
      |             |             |
startTunnel()  stopTunnel()  handleAppMessage()
      |             |             |
[Connect]     [Disconnect]    [App <-> Ext IPC]
      |             |
  sleep()        wake()
      |             |
[Save State]   [Restore State]
```

On-demand activation flow [^64^]:

1. Trigger condition met (Wi-Fi SSID match, domain lookup, etc.)
2. System automatically calls `startTunnel()`
3. Tunnel establishes
4. If condition no longer met, system may call `stopTunnel()`

13.2 Android Lifecycle

```
[App calls VpnService.prepare()] -> [System dialog] ->
[Service.startForeground()]
                                                              |

[VpnService.run() - packet loop]
                                                              |
                                                 [TUN fd readable] ->
[Read packet] -> [Process]
                                                              |
                                                 [Socket readable] ->
[Read packet] -> [Write to TUN]
                                                              |

[Service.stopSelf() / System kills]
```

Always-on lifecycle [^29^]:

1. Device boots
2. System starts VPN service automatically
3. System may restart service if it crashes
4. Service must detect system-initiated vs user-initiated starts

13.3 Windows Lifecycle

For service-based VPN:

```
[Service Auto-Start] -> [Create WinTUN Adapter] -> [WFP Rules
Install]
```

```
                                                    |
                                                [Packet Processing
Loop]
                                                    |
                                                [Session disconnect]
-> [Cleanup]
```

13.4 Reconnection Handling

All platforms require careful reconnection handling:

Platform	Reconnect Mechanism	Key Consideration
iOS	`cancelTunnelWithError()` triggers system retry	Must preserve state in App Group
Android	System restarts service for always-on	Service `onStartCommand()` must handle re-init
Windows	Service control manager	Implement exponential backoff
Linux	systemd auto-restart	`Restart=on-failure` in unit file
HarmonyOS	ExtensionAbility lifecycle	`onCreate()`/ `onDestroy()` patterns

14. Cross-Platform Comparison Matrix

14.1 API × Capability Matrix

Capability	macOS/iOS	Android	Windows	Linux
Custom Protocol (packet-level)	Yes (NEPacketTunnelProvider)	Yes (VpnService TUN)	Yes (WinTUN)	Yes (TU fd)
Custom Protocol (TCP/UDP)	Yes (createUDPSession/ createTCPConnection)	Yes (standard sockets)	Yes (Winsock)	Yes (Be sockets
System-managed persistence	Yes (extension process)	Partial (foreground svc)	Yes (Windows Service)	Yes (sys
Kill switch	PF (macOS) / Limited (iOS)	Always-on + lockdown	WFP / WinTUN auto	nftables
	Yes		Yes	Yes

Capability	macOS/iOS	Android	Windows	Linux
Split tunneling (route)		Yes (API 33+ excludeRoute)		
Split tunneling (per-app)	No (iOS) / Limited (macOS)	Yes (allowed/ disallowed)	Yes (WFP process)	Yes (cg
On-demand rules	Yes (NEOnDemandRule)	No	No	No
MDM configuration	Yes (profile payload)	Yes (AE policies)	Yes (CSP)	Partial
Raw socket access	Limited	No	Yes (admin)	Yes (CAP_N
Battery optimization exempt	N/A	Yes (API whitelist)	N/A	N/A

14.2 API × Limitation Matrix

Limitation	macOS/ iOS	Android	Windows	Linux	HarmonyOS
Single VPN at a time	Yes (enterprise)	Yes	No	No	Yes
Requires user authorization	Yes (system dialog)	Yes (VPN dialog)	No (service)	No (root/ caps)	Yes (permission)
Sandbox restrictions	App Extension model	Foreground service	UWP sandbox	Flatpak/ Snap	Ability sandbox
Memory pressure kills	Yes (extension)	Yes (background)	No (service)	No	Yes (Ability)
Custom protocol kernel code	No	No	Yes (WFP callout)	Yes (kernel module)	No

15. Danger Zones & Platform-Specific Pitfalls

15.1 macOS/iOS Danger Zones

1. **PF routing table conflicts**: Apple explicitly warns against modifying routing tables directly [^136^]:

   > "Avoid using Packet Filter or directly modifying the routing table on the Mac. This is not supported and risk clashing with traffic filtering and routing rules installed by the system or other apps."

2. **utun interface number changes**: After sleep/wake, the interface number (e.g., `utun3` -> `utun4`) can change. Must monitor and update firewall rules.

3. **Enterprise VPN precedence**: Only one enterprise VPN can be active. If multiple VPN apps are installed, they conflict.

4. **Extension memory limits**: System may terminate the packet tunnel provider under memory pressure. Must be efficient.

5. **iOS Simulator doesn't support Network Extensions**: Must test on physical hardware.

6. **App Store rejection risk**: VPN apps face heightened scrutiny. Must have clear privacy policy and legitimate use case.

15.2 Android Danger Zones

1. **Foreground service 6-hour limit (Android 15)**: Critical for always-on VPN. Must migrate to UIDT jobs [^177^].

2. **Battery optimization**: OEM-specific "battery saver" features (Samsung, Xiaomi, etc.) can kill VPN services despite system settings.

3. **VPN permission revocation**: If user revokes VPN permission in Settings, the service is killed without warning.

4. **Multiple VPN apps**: Only one VPN can be active. Starting another VPN silently stops the current one.

5. **HttpProxy over split tunnel**: Using `setHttpProxy()` with split tunneling "generally won't work as expected" [^109^].

6. **excludeRoute() API 33+ only**: Must maintain dual code paths for route exclusion.

7. **Private Space (Android 15+)**: VPN doesn't automatically apply to Private Space apps [^26^].

15.3 Windows Danger Zones

1. **BFE service restart**: If the Base Filtering Engine service restarts, WFP rules are lost. Must monitor and restore.

2. **HVCI compatibility**: Kernel-mode drivers must be compatible with Hypervisor-Protected Code Integrity on modern systems.

3. **Interface metric race**: Windows may override VPN interface metric, causing traffic to bypass the tunnel.

4. **Multiple filter drivers**: Conflicts between antivirus, VPN, and other WFP callout drivers are common.

5. **TDI filter deprecation**: Transport Driver Interface filters are deprecated; must use WFP.

6. **WireGuard kill-switch lockout**: Using `AllowedIPs = 0.0.0.0/0` with the kill switch can block LAN access, including router admin interfaces [^151^].

15.4 Linux Danger Zones

1. **iptables vs nftables fragmentation**: Must support both or target specific distributions.

2. **DNS leak**: Linux doesn't automatically manage DNS when VPN connects. Must configure systemd-resolved or update resolv.conf.

3. **Flatpak/Snap sandbox**: TUN device access is blocked in sandboxed apps. Requires classic confinement or external daemon.

4. **Capability management**: `CAP_NET_ADMIN` must be set on the binary; easy to misconfigure.

5. **Route table conflicts**: Multiple VPN clients or network management tools can conflict on route table entries.

6. **WireGuard kernel module availability**: Not all distributions have the kernel module (especially older kernels).

15.5 HarmonyOS Danger Zones

1. **System API restriction**: `MANAGE_VPN` was restricted to system apps until API 12. Third-party support is still maturing.

2. **ACL requirement**: Apps need Access Control List configuration for VPN permission.

3. **Native C interop**: ArkTS/ArkUI apps need NAPI for Rust integration, adding complexity.

4. **Ecosystem maturity**: VPN ecosystem on HarmonyOS is less mature than Android/iOS.

15.6 Aurora OS Danger Zones

1. **Userspace WireGuard only**: No kernel module; uses `wireguard-go` with performance implications.

2. **Community package dependency**: Relies on OpenRepos for VPN packages, not official repositories.

3. **ConnMan stability**: Reports of VPN plugin crashes and connection issues.

4. **Limited documentation**: Mostly community-maintained documentation.

---

16. Recommendations for Rust Shared Core Design

16.1 Platform Adapter Architecture

Based on this research, the Rust shared core should implement a **platform adapter pattern** with the following interfaces:

```rust
// Core abstraction for packet I/O
trait TunnelDevice {
    fn read_packet(&mut self, buf: &mut [u8]) -> Result<usize>;
    fn write_packet(&mut self, packet: &[u8]) -> Result<()>;
    fn set_mtu(&mut self, mtu: u16) -> Result<()>;
}

// Platform-specific VPN service lifecycle
trait PlatformAdapter {
    fn setup_tunnel(&self, config: TunnelConfig) ->
        Result<Box<dyn TunnelDevice>>;
    fn protect_socket(&self, socket: RawFd) -> Result<()>;
    fn configure_routes(&self, routes: &[Route]) -> Result<()>;
    fn configure_dns(&self, servers: &[IpAddr]) -> Result<()>;
    fn enable_kill_switch(&self) -> Result<()>;
    fn disable_kill_switch(&self) -> Result<()>;
```

```
    fn set_split_tunnel_apps(&self, allowed: &[String], blocked:
        &[String]) -> Result<()>;
}
```

16.2 Key Design Decisions

1. **Accept TUN fd from platform**: All platforms provide a TUN file descriptor. The Rust core should accept this fd and perform packet I/O through it.

2. **Separate transport from tunnel**: The platform adapter should handle the UDP/TCP transport socket to the VPN server (including `protect()`), while the Rust core handles packet encryption/decryption.

3. **Platform manages routing**: Route configuration should be handled by the native platform code (NetworkExtension on Apple, VpnService.Builder on Android, netlink on Linux, etc.).

4. **Kill switch delegated to platform**: Each platform has its own optimal kill switch mechanism. The Rust core should expose `enable/disable_kill_switch()` but the implementation should be platform-native.

5. **WireGuard in kernel space (Linux)**: On Linux, use the kernel WireGuard module via Netlink when available, falling back to userspace implementation.

6. **Foreground service handling (Android)**: The Android adapter must properly manage the foreground service lifecycle, notification, and Android 15 UIDT job migration.

7. **App Groups for state persistence (iOS)**: Store connection state in App Group shared container for recovery after extension termination.

16.3 Implementation Priority

Priority	Platform	Rationale
P0	Linux	Most flexible, easiest to prototype
P0	Android	Largest mobile market, well-documented API
P0	macOS	NetworkExtension is mature and well-supported
P1	Windows	Large desktop market, WFP complexity manageable
P1	iOS	Similar to macOS but with additional constraints
P2	HarmonyOS	Emerging market, API still maturing

Priority	Platform	Rationale
P2	Aurora OS	Niche market, community-driven

Source Index

[^21^] Stack Overflow — "How to implement VPN split tunneling in Android's VpnService" (2024)
[^25^] Android Source — `VpnService.java` implementation (AOSP main)
[^26^] Jason Bayton — "Is it possible to utilise a single VPN connection across the entire device?" (2026)
[^29^] Android Developer Documentation — "VPN | Connectivity" (2026)
[^63^] Samsung Knox — "Apps not working when the device is in battery optimization mode"
[^64^] Anton Gubarenko — "iOS Network Extensions and Personal VPN: A Developer's Guide" (2025)
[^65^] Kean.blog — "VPN, Part 1: VPN Profiles" (2020)
[^66^] NT Kernel — "Windows Packet Filter" (2024)
[^67^] Thomas Haller — "WireGuard in NetworkManager" (2019)
[^69^] NetworkSpy — "iOS Packet Tunnel Provider Code Walkthrough"
[^70^] Newly — "How to Get the Apple Network Extension (VPN) Entitlement" (2026)
[^71^] KaOS Forum — "Wireguard VPN?" discussion (2021)
[^73^] OSR Developer Community — "NDIS Intermediate Driver or WFP?" (2012)
[^77^] Huawei Developer — "@ohos.app.ability.VpnExtensionAbility" (2026)
[^107^] ENEBA — "What Is a VPN Kill Switch and Why You Need One" (2026)
[^108^] Tailscale GitHub — "Feature request: Kill switch / block traffic when exit node is unreachable" (2026)
[^109^] Android Developer Reference — `VpnService.Builder` (2026)
[^111^] Mullvad — "Split tunneling with the Mullvad app" (2026)
[^112^] Proton VPN — "How to use split tunneling"
[^114^] IVPN — "Do you offer a kill switch or VPN firewall?"
[^117^] Stack Exchange Security — "How do VPN kill switches in mobile apps actually work?" (2021)
[^129^] Kean.blog — "VPN, Part 2: Packet Tunnel Provider" (2020)
[^136^] Apple WWDC 2025 — "Filter and tunnel network traffic with NetworkExtension"
[^142^] Juejin — "NetworkExtension Tunnel Development Complete Code" (2020)
[^143^] Sid Shanker — "Using Linux Raw Sockets" (2018)
[^145^] GitHub — "WireGuard VPN plugin for Sailfish OS"

[^147^] Linux man-pages — `capabilities(7)`
[^148^] Apple Developer — `disconnectOnSleep` documentation
[^149^] Nico Cartron — "SailfishOS as WireGuard endpoint"
[^151^] SNB Forums — "Wireguard VPN Client: killswitch activation -> LAN administration lock-out" (2024)
[^152^] SailfishOS Wiki — "Installing WireGuard"
[^154^] SailfishOS Forum — "Wireguard in SailfishOS 5.0" (2025)
[^156^] WireGuard Windows — "Network Configuration Quirks"
[^170^] Android Developers Blog — "Modern background execution in Android" (2018)
[^173^] Huawei Central — "HarmonyOS 4.1 release beta 1 with API 11 interfaces" (2024)
[^174^] Huawei Developer — "New and Enhanced Features - HarmonyOS 6.0.0" (2026)
[^175^] Roberto Huertas — "Building an Android service that never stops running" (2019)
[^177^] Chaitanyaduse Medium — "Long-Running Background Work in Android and Its Quirks" (2025)
[^178^] Android Developer — "Support for long-running workers"
[^193^] ViaSocket — "7 Best Device Provisioning Platforms for IT Teams" (2026)
[^195^] Seaxiang Blog — "@ohos.net.vpn (VPN Management)" (2023)
[^197^] OpenHarmony Docs — "Permissions for System Applications"
[^199^] Microsoft GitHub — "UwpVpnPluginSample"
[^219^] VPN.how — "How the OS Kernel Blocks Traffic and Protects Your Privacy" (2026)
[^220^] GrapheneOS — "Add configuration option to add exceptions to Block connections without VPN mode" (2025)
[^221^] Hacker News — Kill switch implementation discussion (2024)
[^224^] GitHub — "split-tunnel: callout driver implementing policy-based routing for Windows"

Research compiled from 12 independent web searches across 60+ authoritative sources including official developer documentation, open-source code repositories, security research papers, and industry technical blogs. All citations use [^number^] format with source URLs preserved.